

# ELI — Adaptive Inference Governor

ELI v2.4.67

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## Adaptive Inference Governor — Status and Remaining Plan

**Updated for v2.4.67.** Most of the gaps this plan identified in June are closed; the two that need a governor (throughput-aware think decision and per-call wall-time projection) are still open. Status per gap below; no governor code exists yet.

**Scope:** make ELI’s reasoning and inference *policy* (think or no-think, token budget, pass count, recovery) a pure function of the runtime environment and the task, so one redistributed binary behaves correctly on any machine with any model and no per-machine tuning.

**Origin:** a 2026-06-17 session in which EXPLAIN\_COGNITION\_RUNTIME burned a 273 s empty compact-synthesis pass, then two 539 s empty standard-synthesis passes, before a two-stage chain-of-thought fallback (470 s and 502 s): about 38 minutes for one question. The working final answer was 4,071 characters (about 1,018 tokens) in 502 s, roughly 2 tokens per second.

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### 0. The one-sentence problem

ELI adapts its inference to **memory** (VRAM, context length) and **model size**, but not to **measured compute speed** (tokens per second). On a slow machine the same prompt that works on the development box can produce multi-minute turns.

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### 1. What ELI already adapts to (do not rebuild)

| Capability                                  | Where                                                                                               | What it adapts to                                                                       |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Model-agnostic resolution (no baked model)  | gguf_inference.get_model_path                                                                       | settings, environment                                                                   |
| Chat family and template                    | cognition/model_identity.py, model_output_tokens.py                                                 | GGUF metadata (template, architecture), not the file name                               |
| Thinking-channel detection                  | gguf_inference._is_thinking_model → model_identity.is_thinking_model                                | the embedded chat template, then architecture; no name list                             |
| Trained context length                      | core/startup_hardware_optimizer, ELI_MODEL_TRAINING                                                 | ELI_MODEL_TRAINING, then real GGUF context_length, else 8192; no filename table         |
| Free-VRAM-aware sizing, one fit calculation | core/hardware_profile.py (smart_fit_config, unified_fit_config)                                     | free VRAM, RAM budget, fit priority                                                     |
| KV-cache and compute-buffer reserve         | hardware_profile.py (_kv_cache_mb, _compute_graph_reserve_mb)                                       | context, layers, batch                                                                  |
| Graceful GPU-layer fallback                 | boot optimiser; runtime_snapshot.json                                                               | load success                                                                            |
| Per-mode token budget                       | adaptive_load_report cognition/reasoning_modes.py                                                   | n_ctx (20 % of the window for quick, 30 % otherwise), prompt pressure, complexity, mode |
| Size tier → budget scale                    | detect_tier, tier_scale                                                                             | model size                                                                              |
| Live decode-speed EMA                       | model_tier.record_speed, measured_tok_s (fed from the non-stream generation path in gguf_inference) | measured tokens per second                                                              |
| Speed-capped pass count                     | model_tier.speed_passes                                                                             | measured tokens per second (thresholds ELI_SLOW_TPS=5, ELI_FAST_TPS=15)                 |
| No-think on utility calls                   | gguf_inference._no_think_prefill, force_no_think()                                                  | structured output, small budgets (< 1024), quick mode                                   |
| Redistribution-aware settings               | core/runtime_settings.py                                                                            | machine moves                                                                           |

The speed EMA is consumed in exactly one place, speed\_passes(). Nothing in the think decision, the token budget or mode selection reads it. speed\_passes() reduces how many full generations a mode runs and never caps output length.

## 2. The gaps and where each stands

| Gap                                                             | Was                                                | Status                                                                                                                     |
|-----------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>A</b> thinking detection was a filename allowlist            | _is_thinking_model matched qwen3, deepseek-r1, ... | <b>Closed.</b> Detection reads the chat template, then the architecture.                                                   |
| <b>B</b> the think decision and token budget ignored throughput | no tok/s term in _no_think_prefill or the budget   | <b>Open.</b> The inputs are structured, max_tokens < 1024, the quick mode, the force_no_think() scope and ELI_MODEL_THINK. |

| Gap                                                    | Was                                                                | Status                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>C</b> no wall-time projection                       | nothing computes “at the live EMA this call takes about N seconds” | <b>Open.</b>                                                                                                                                                                                                                                                                                                                                              |
| <b>D</b> empty output was not self-correcting          | an unterminated <think> returned empty, then a full-price retry    | <b>Partly closed.</b> An empty non-stream response is retried once with <code>force_no_think</code> in the engine ( <code>_get_chat_response</code> ), and an empty stream is retried once with <code>no-think</code> in <code>gguf_inference</code> . There is still no reserved answer budget.                                                          |
| <b>E</b> context sizing undershot the trained context  | a filename table returned 32768 for <code>qwen3</code>             | <b>Closed.</b> GGUF metadata is authoritative.                                                                                                                                                                                                                                                                                                            |
| <b>F</b> <code>force_no_think</code> was a point patch | one call site                                                      | <b>Partly closed.</b> It now wraps compact grounded synthesis, the chain-of-thought stage prose, the correction repair, and the empty-response and stream retries in <code>_get_chat_response</code> . The standard synthesis fallback ( <code>_synthesize_answer</code> ) and coding-agent fix generation for <code>SELF_IMPROVE</code> are not wrapped. |
| <b>G</b> redundant proactive churn                     | <code>persona_updater</code> rebuilt the same overlay every tick   | <b>Closed.</b> Persona and user-profile overlays are debounced (120 s) with a content dirty-check.                                                                                                                                                                                                                                                        |

Note on gap G: `persona_updater` is deterministic (database reads and regexes); the cost was redundant CPU and SQLite work, not inference-lock contention.

### 3. Design for what remains: the Adaptive Inference Governor

A single policy object that every model call routes its *decision* through. It consumes `reasoning_modes`, `model_tier` and `hardware_profile`; it does not replace them.

#### 3.1 Inputs (all available at runtime)

- **Model capability:** `model_identity.is_thinking_model(template, architecture)`, cached per model.
- **Throughput:** `model_tier.measured_tok_s()`, with a cold-start default from `detect_tier()` until the first measurement.
- **Context:** `n_ctx` from the runtime snapshot and `n_ctx_train` from metadata.
- **Task class:** below.

#### 3.2 Task classes

1. **STRUCTURED:** JSON, routing, plan graphs. Never thinks (already true via `structured=True`).
2. **GROUNDING\_SYNTHESIS:** evidence already gathered; the call only phrases it (the `EXPLAIN_*` actions, RAG answer synthesis, fix-patch rendering). Thinking adds nothing; force no-think.
3. **OPEN\_REASONING:** genuine multi-step reasoning (chain of thought, tree of thoughts, constitutional, self-consistency, open chat on a hard question). Thinking is allowed but bounded by throughput.

### 3.3 The decision (a pure function)

```
policy(task_class, model_caps, tok_s, n_ctx, prompt_tokens) -> {  
  think: bool,  
  think_budget: int,      # max tokens allowed inside <think>  
  answer_budget: int,     # reserved tokens after </think>  
  passes: int,            # via speed_passes()  
}
```

- A model that does not think: think=False always.
- STRUCTURED or GROUNDED\_SYNTHESIS: think=False.
- OPEN\_REASONING: at or above ELI\_FAST\_TPS think with the full budget; between slow and fast scale think\_budget so the projected wall time is within the target; at or below ELI\_SLOW\_TPS turn thinking off, or hard-cap think\_budget.
- Always reserve answer\_budget, so a thinking call cannot exhaust its budget before it answers.

### 3.4 Integration points

- gguf\_inference.\_no\_think\_prefill: consult the governor instead of only structured, < 1024 and the environment; keep force\_no\_think() as the primitive and ELI\_MODEL\_THINK as the hard user override.
- reasoning\_modes: split the target tokens into think\_budget and answer\_budget.
- Engine synthesis sites: tag\_compact\_grounded\_synthesis and \_synthesize\_answer as GROUNDED\_SYNTHESIS and the reasoning-mode runners as OPEN\_REASONING.
- Feed the speed EMA from the stream path as well, so quick-mode chat updates it.

### 3.5 Configuration

ELI\_SLOW\_TPS and ELI\_FAST\_TPS exist (5 and 15). A soft per-turn wall-time target (ELI\_TURN\_LATENCY\_TARGET\_S) is proposed and does not exist yet. ELI\_MODEL\_THINK is the hard override. Nothing machine-specific is persisted.

## 4. Cross-machine behaviour to aim for

| Machine and model                              | tok/s    | grounded actions<br>(EXPLAIN_*) | hard chat                                              |
|------------------------------------------------|----------|---------------------------------|--------------------------------------------------------|
| 8 GB GPU with a 30B MoE                        | about 2  | no-think, one pass              | bounded think, within the target, never empty          |
| 24 GB GPU with a 7B                            | about 40 | no-think                        | full think, full budget                                |
| CPU-only with a 7B                             | about 3  | no-think                        | think off or<br>hard-capped, with an<br>answer reserve |
| any machine with a non-thinking model          | n/a      | normal answer                   | normal answer                                          |
| any machine with an unfamiliar reasoning model | any      | detected through the template   | handled                                                |

## 5. Phases

1. **Governor core** for grounded synthesis: the policy function, no-think wired through \_no\_think\_prefill, coding-agent fix rendering tagged.
2. **Throughput-bounded open reasoning** with an answer reserve: split budgets in reasoning\_modes.

3. **EMA from streaming** so quick mode updates the measured speed.

Each phase is independently shippable.

## 6. Verification

- Unit: `policy(...)` truth table over task class  $\times$  throughput tier  $\times$  model-thinks.
- No-model engine tests in the style of `tests/test*_no_gguf*.py`.
- Behavioural: `EXPLAIN_MEMORY_RUNTIME` and `EXPLAIN_COGNITION_RUNTIME` synthesis is non-empty on the first pass and runs no-think.
- Speed simulation: inject a fake `measured_tok_s()` and assert the think flag and budget flip.
- Existing guards that must stay green: `tests/test_gguf_think_stop_collision.py`, `tests/test_reasoning_mode_collision.py`, `tests/test_cot_think_budget.py`.

## 7. Risks and non-goals

- **Over-suppressing thinking** on hard turns at mid speed: only `GROUNDED_SYNTHESIS` is unconditionally no-think; `OPEN_REASONING` keeps thinking with a budget above slow.
- **Non-goals:** changing model selection, the agent set, persona content or the grounding contract; removing `ELI_MODEL_THINK`.